

Ashik Ahamed

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Research Interests

Machine Learning & Deep Learning · Time-Series Forecasting · Explainable Artificial Intelligence (XAI) · Climate Data Analytics · Feature Selection Methods · Statistical Modeling · Predictive Analytics

Education

American International University–Bangladesh (AIUB)

2021 – 2025

B.Sc. in Computer Science, Major: Information Systems

Dhaka, Bangladesh

- CGPA: 3.89 / 4.00 (Dean's List – 4 times)
- Relevant Coursework: Machine Learning, Data Science, Probability & Statistics, Data Warehousing & Data Mining, Computer Vision & Pattern Recognition, Artificial Intelligence & Expert Systems

Publications & Ongoing Research

Enhancing Heatwave Prediction Using Explainable AI: Feature Selection and Model Optimization for Improved Forecasting

Ahamed, A. *, et al. — *International Conference on Frontiers in Computing and Systems (COMSYS 2025)* — Accepted, in press

- Collected and preprocessed a 10-year climate dataset (2010–2020) for Dhaka
- Applied SHAP-based feature selection to identify key predictors of daily maximum temperature
- Implemented and benchmarked GRU, LSTM, RNN, and XGBoost architectures
- Evaluated models using MSE, MAE, and R^2

Time-Series Analysis of Climate Extremes: A Multi-City Heatwave Prediction and Explainability Study

May 2026 – Present

In collaboration with Nuzhat Tabassum, Assistant Professor, AIUB

- Designed a three-phase pipeline (feature-based prediction, pure time-series forecasting, explainability) on an 14-year (2010–2024) daily temperature dataset for Dhaka
- Benchmarked nine feature-selection-driven ML models (Decision Tree, Random Forest, XGBoost, AdaBoost, CatBoost, LightGBM, Stacking, Blending, Metaheuristic/GA); champion Stacking model achieved MAE 0.66°C, R^2 0.959
- Benchmarked classical and deep time-series models (ARIMA, SARIMA, LSTM, BiLSTM, RNN, GRU, CNN-LSTM, PyCaret); champion GRU model achieved R^2 0.78
- Applied LIME (local) and Morris Sensitivity Analysis (global) to explain champion models, identifying dry-bulb temperature as the dominant physical driver
- Extended the pipeline to three additional cities (Rajshahi, Chittagong, Sylhet) spanning distinct climate zones to evaluate cross-city generalizability, revealing strong regression generalization (R^2 0.68–0.98) but city-dependent classification performance (F1 0.00–0.88)

Machine Learning Framework for Predicting Daily Heat-Related Illness Risk Across Bangladesh

June 2026 – Present

Independent Research Project

- Building an ML pipeline on 11 years of weather data from 8 cities across Bangladesh to predict daily heat-related illness risk (heatstroke, heat exhaustion) rather than temperature alone
- Deriving four risk categories (Safe, Caution, Danger, Extreme Danger) from a Heat Index computed using temperature and humidity
- Benchmarking XGBoost, Random Forest, and LSTM for multi-class risk-level prediction
- Applying SHAP to identify the dominant city-specific risk factors and proposing an early-warning framework
- Positions itself as the first Bangladesh-focused study to model health risk directly, as prior work has predicted temperature only

Research Experience

Research Student

Sep 2026 – Present

Elite Research Lab

- Receiving structured mentorship on academic paper writing, including section-wise manuscript structure, framing research contributions, and positioning related work, with a focus on producing publication-ready, A*-venue-quality manuscripts
- Concurrently co-authoring an ongoing research paper under lab supervision, applying the writing methodology in parallel with training

Research Assistant

May 2025 – March 2026

Supervised by Nuzhat Tabassum, Assistant Professor, AIUB

Dhaka, Bangladesh

- Performed data cleaning and preprocessing to ensure high-quality datasets for climate research
- Implemented and evaluated machine learning models to support ongoing research outcomes
- Prepared technical documents in L^AT_EX and presentations to communicate research findings

Professional Experience (Non-Research)

Content Developer, General R&D Department

April 2026 – Present

Lecture Publications

Dhaka, Bangladesh

- Develop academic book content across subjects, ensuring conceptual accuracy, depth, and alignment with curriculum standards
- Simplify and restructure complex academic material into clear, pedagogically accessible explanations for students
- Research topics in depth to ensure factual accuracy and relevance of content before publication

Academic Content Writer

March 2026 – Present

International Research Collaboration Bangladesh

Dhaka, Bangladesh

- Produced academic content, including research articles and thesis-related documents, based on client requirements
- Structured and formatted documents following specific guidelines (word count, tone, referencing style)

Data Analyst

Sep 2025 – Feb 2026

DC IT Management

Dhaka, Bangladesh

- Analyzed transaction records to identify seasonal sales patterns and support strategic decisions; reduced monthly reporting time by 30% through automated SQL–Python pipelines
- Built Power BI dashboards adopted by sales department managers for ongoing monitoring

Junior Data Analyst

Dec 2024 – April 2025

Super Star Group

Dhaka, Bangladesh

- Applied machine learning models to business data to identify trends and support supervisor decision-making

Selected Projects

Explainable AI in Education for Personalized Learning

2024 – 2025

Python, Scikit-learn, XGBoost, SHAP

- Developed an XAI framework to predict student GPA and classify stress levels from a lifestyle dataset; applied SHAP for global/local interpretability, identifying study hours, sleep, and physical activity as the most influential predictors (Linear Regression $R^2=0.538$; SVC 95% accuracy, F1=0.94)

Exploring Data Visualization in Pharmaceutical Evaluation

R, Statistical Analysis, Data Visualization

- Conducted exploratory data analysis on pharmaceutical datasets and developed publication-quality visualizations using ggplot2 to identify significant variables influencing drug-evaluation outcomes

Face Recognition System

Python, TensorFlow, Deep Learning

- Designed and implemented a CNN-based face recognition system; preprocessed and augmented image data and evaluated performance via accuracy, precision, recall, and confusion matrix analysis

Technical Skills

- **Programming:** Python, R, SQL
- **ML & Deep Learning:** Scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, PyTorch
- **Time-Series & Forecasting:** ARIMA, SARIMA, LSTM, GRU, BiLSTM, CNN-LSTM, PyCaret
- **Explainability:** SHAP, LIME, Morris Sensitivity Analysis
- **Data Tools:** Power BI, MySQL, Jupyter Notebook, L^AT_EX

Awards & Honors

- Conference Presenter, International Conference on Frontiers in Computing and Systems (COMSYS 2025)
- Dean's List, AIUB — awarded four times for outstanding academic performance (2021–2025)
- IELTS Academic — Overall Band 7.0 (C1 level), 2026
- ICPC Asia Dhaka Regional — Online Preliminary Contest participant, Team AIUB Virtuosos, Oct 2023

References

Nuzhat Tabassum
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